

Homework 9

Solution

- For each of the following series, find the domain of convergence and compute its value (as a function of x).

(a)

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{2^n \cdot 2} \cdot (-2x + 2)^{2n+2}$$

Solution:

We write

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-2)^n}{2^n \cdot 2} \cdot (-2x + 2)^{2n+2} &\stackrel{t=(-2x+2)^2}{=} \sum_{n=0}^{\infty} \frac{(-2)^n}{2^n \cdot 2} \cdot t^{n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 2^n}{2^n \cdot 2} \cdot t^{n+1} \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} (-1)^{n+1} t^{n+1} = -\frac{1}{2} \sum_{n=0}^{\infty} (-t)^{n+1} \stackrel{u=-t}{=} -\frac{u}{2} \sum_{n=0}^{\infty} u^n \end{aligned}$$

Thus, the series converges if and only if

$$|u| < 1 \Leftrightarrow |(-2x + 2)^2| < 1 \Leftrightarrow \frac{1}{2} < x < \frac{3}{2}$$

and for every $\frac{1}{2} < x < \frac{3}{2}$ we have

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-2)^n}{2^n \cdot 2} \cdot (-2x + 2)^{2n+2} &= -\frac{u}{2} \sum_{n=0}^{\infty} u^n = -\frac{u}{2} \cdot \frac{1}{1-u} \\ &= \frac{t}{2} \cdot \frac{1}{1+t} = \frac{(2-2x)^2}{2} \cdot \frac{1}{1+(2-2x)^2} \end{aligned}$$

(b)

$$\sum_{n=0}^{\infty} (-2)^n \frac{2^n \cdot 2}{(-2x+2)^{2n+2}}$$

Solution:

We have

$$\begin{aligned} \sum_{n=0}^{\infty} (-2)^n \frac{2^n \cdot 2}{(-2x+2)^{2n+2}} &= \sum_{n=0}^{\infty} (-2)^n \frac{2^n \cdot 2}{\left((-2x+2)^2\right)^n \cdot (-2x+2)^2} \\ &= \frac{2}{(-2x+2)^2} \sum_{n=0}^{\infty} \left(-\frac{4}{(-2x+2)^2}\right)^n \stackrel{t = -\frac{4}{(-2x+2)^2}}{=} -\frac{t}{2} \sum_{n=0}^{\infty} t^n \end{aligned}$$

Thus the series converges iff

$$|t| < 1 \Leftrightarrow \frac{4}{(-2x+2)^2} < 1 \Leftrightarrow x < 0 \text{ or } x > 2$$

Thus for every $x < 0$ or $x > 2$ we have

$$\sum_{n=0}^{\infty} (-2)^n \frac{2^n \cdot 2}{(-2x+2)^{2n+2}} = -\frac{t}{2} \sum_{n=0}^{\infty} t^n = -\frac{t}{2} \cdot \frac{1}{1-t} = \frac{2}{(-2x+2)^2} \cdot \frac{1}{1 + \frac{4}{(-2x+2)^2}}$$

2. Find the domain of convergence for each of the following power series:

(a)

$$\sum_{n=1}^{\infty} \frac{n}{5^n} \cdot x^n$$

Solution:

First we find the radius of convergence

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n}{5^n}}} = 5$$

Thus the series converges for every $-5 < x < 5$.

Now we need to check if the series converges for $x = 5, -5$.

For $x = 5$:

We get the series

$$\sum_{n=1}^{\infty} \frac{n}{5^n} \cdot 5^n = \sum_{n=1}^{\infty} n$$

As $\lim_{n \rightarrow \infty} n = \infty \neq 0$, then the series diverges since the necessary criterion does not hold.

For $x = -5$:

We get the series

$$\sum_{n=1}^{\infty} \frac{n}{5^n} \cdot (-5)^n = \sum_{n=1}^{\infty} (-1)^n n$$

As $\lim_{n \rightarrow \infty} [(-1)^{2n} 2n] = \lim_{n \rightarrow \infty} (2n) = \infty \neq 0$, then $\lim_{n \rightarrow \infty} [(-1)^n n] \neq 0$, and therefore the series diverges since the necessary criterion does not hold.

Thus, the domain of convergence is $-5 < x < 5$.

(b)

$$\sum_{n=0}^{\infty} \frac{3^n}{(2n+1)!} \cdot (x-1)^{2n}$$

Solution:

First, we find the radius of convergence

$$\sum_{n=0}^{\infty} \frac{3^n}{(2n+1)!} (x-1)^{2n} \underset{t=(x-1)^2}{=} \sum_{n=0}^{\infty} \frac{3^n}{(2n+1)!} t^n$$

We note that

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{3^{n+1}}{(2n+3)!}\right)}{\left(\frac{3^n}{(2n+1)!}\right)} = \lim_{n \rightarrow \infty} \frac{3}{(2n+2)(2n+3)} = 0$$

Thus $R = \infty$.

Thus, the series converges for every $t \in \mathbb{R}$, and therefore the series converges for every $x \in \mathbb{R}$.

(c)

$$\sum_{n=1}^{\infty} \frac{3^n + 5^n}{n} \cdot x^n$$

Solution:

First, we find the radius of convergence

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\frac{3^n + 5^n}{n}}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n}}{\sqrt[n]{5^n (1 + (\frac{3}{5})^n)}} = \frac{1}{5}$$

Thus the series converges for every $-\frac{1}{5} < x < \frac{1}{5}$.

Now we need to check if the series converges for $x = \frac{1}{5}, -\frac{1}{5}$.

For $x = \frac{1}{5}$:

We have

$$\sum_{n=1}^{\infty} \frac{3^n + 5^n}{n} \cdot \left(\frac{1}{5}\right)^n = \sum_{n=1}^{\infty} \frac{\left(\frac{3}{5}\right)^n + 1}{n} \geq \sum_{n=1}^{\infty} \frac{1}{n} \stackrel{\text{Harmonic}}{=} \infty$$

Thus the series $\sum_{n=1}^{\infty} \frac{(\frac{3}{5})^n + 1}{n}$ is divergent by the comparison test.

For $x = -\frac{1}{5}$:

We get the series

$$\sum_{n=1}^{\infty} \frac{3^n + 5^n}{n} \cdot \left(-\frac{1}{5}\right)^n = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{(\frac{3}{5})^n + 1}{n} = - \sum_{n=1}^{\infty} (-1)^{n+1} \cdot \left[\frac{1}{n} \cdot \left(\frac{3}{5}\right)^n + \frac{1}{n} \right]$$

Note that the series $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{1}{n} \cdot \left(\frac{3}{5}\right)^n$ converges absolutely as

$$\sum_{n=1}^{\infty} \frac{1}{n} \cdot \left(\frac{3}{5}\right)^n \leq \sum_{n=1}^{\infty} \left(\frac{3}{5}\right)^n \stackrel{\text{tail of geometric}}{<} \infty$$

Thus,

$$- \sum_{n=1}^{\infty} (-1)^{n+1} \cdot \left[\frac{1}{n} \cdot \left(\frac{3}{5}\right)^n + \frac{1}{n} \right] = \text{convergent} + \text{Leibniz} = \text{convergent}$$

Thus, the domain of convergence is $-\frac{1}{5} \leq x < \frac{1}{5}$.

3. Find the domain of convergence of the power series:

$$\sum_{n=1}^{\infty} \frac{x^{2^n}}{n^2} = x^2 + \frac{x^4}{4} + \frac{x^8}{9} + \frac{x^{16}}{16} + \dots$$

Solution:

We first note that the series is convergent for $x = 0$. We now need to check convergence for every $x \neq 0$.

Let $0 \neq x \in \mathbb{R}$. Note that the series is positive, and we can use the ratio test. We compute:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\left[\frac{x^{2^{n+1}}}{(n+1)^2} \right]}{\left[\frac{x^{2^n}}{n^2} \right]} = \lim_{n \rightarrow \infty} \frac{(x^{2^n})^2 \cdot n^2}{x^{2^n} (n+1)^2} = \lim_{n \rightarrow \infty} \frac{x^{2^n} \cdot n^2}{(n+1)^2} = \lim_{n \rightarrow \infty} \frac{x^{2^n}}{\left(1 + \frac{1}{n}\right)^2} = \begin{cases} 0 & |x| < 1 \\ \infty & |x| > 1 \end{cases}$$

It follows that the series is convergent for $|x| < 1$ by the ratio test.

For $|x| > 1$, we obtain $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \infty$. We show that in this case the series is divergent:

As $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \infty$, there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$\frac{a_{n+1}}{a_n} > \boxed{1} \Rightarrow a_{n+1} \geq a_n$$

It follows that the sequence $(a_n)_{n=N_1}^{\infty}$ is increasing, and therefore $a_n \geq a_{N_1}$ for every $n \geq N_1$.

We now prove that this implies $\lim_{n \rightarrow \infty} a_n \neq 0$:

Assume towards a contradiction that $\lim_{n \rightarrow \infty} a_n = 0$. By monotonicity of limits,

$$0 = \lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} a_{N_1} = a_{N_1} > 0$$

Contradiction.

We have proved that for $|x| > 1$ it holds that $\lim_{n \rightarrow \infty} a_n \neq 0$, so the series is divergent by the necessary criterion.

We now need to check convergence for the case $|x| = 1$, i.e., for $x = \pm 1$.

Note that in this case

$$\sum_{n=1}^{\infty} \frac{x^{2^n}}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} \stackrel{2\text{-Harmonic}}{<} \infty$$

We conclude that the domain of convergence of the series is $[-1, 1]$.

4. Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers and let $0 \leq R \leq \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n x^n$. Prove or disprove each of the following statements:

- (a) If $\sum_{n=0}^{\infty} a_n (1.1)^n$ is convergent, then $\sum_{n=0}^{\infty} (-1)^{n+1} a_n$ converges absolutely.

Solution:

True.

As $\sum_{n=0}^{\infty} a_n (1.1)^n$ is convergent, then $R \geq 1.1$ and therefore the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for every $|x| < 1.1$.

Thus, in particular, the series converges absolutely for $x = -1$.

Thus

$$\sum_{n=0}^{\infty} |(-1)^{n+1} a_n| = \sum_{n=0}^{\infty} |a_n (-1)^n| < \infty$$

- (b) If $R < 1$, then $\lim_{n \rightarrow \infty} a_n \neq 0$.

Solution:

True.

ATC that $\lim_{n \rightarrow \infty} a_n = 0$. Thus there exists an $N \in \mathbb{N}$ such that for every $n \geq N$ we have

$$|a_n - 0| < \boxed{1} \Rightarrow |a_n| < 1$$

We know that $\sum_{n=N}^{\infty} x^n$ converges absolutely iff $-1 < x < 1$.

Let $-1 < x_1 < 1$. We note that

$$\sum_{n=N}^{\infty} |a_n x_1^n| \leq \sum_{n=N}^{\infty} |x_1^n|$$

Thus the series $\sum_{n=N}^{\infty} a_n x^n$ converges absolutely at x_1 .

Thus the series $\sum_{n=N}^{\infty} a_n x^n$ converges absolutely for every $-1 < x < 1$ and therefore the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for every $-1 < x < 1$, and therefore $R \geq 1$. Contradiction!

(c) If $R < 1$, then $\lim_{n \rightarrow \infty} |a_n| = \infty$.

Solution:

False.

Counter-example: we define for every $0 \leq n \in \mathbb{Z}$

$$a_n = \begin{cases} 2^k & n = 2k \\ 0 & n = 2k - 1 \end{cases} = \begin{cases} 2^{\frac{n}{2}} & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}$$

Thus

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{k=0}^{\infty} 2^k x^{2k} \stackrel{t=x^2}{=} \sum_{k=0}^{\infty} 2^k t^k$$

We compute the radius of the power series $\sum_{k=0}^{\infty} 2^k t^k$,

$$R_1 = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{2^n}} = \frac{1}{2}$$

Thus the series converges for every

$$|t| < \frac{1}{2} \Leftrightarrow |x^2| < \frac{1}{2} \Leftrightarrow -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

Thus the radius of the series with x is $R = \frac{1}{\sqrt{2}} < 1$.

But $\lim_{n \rightarrow \infty} |a_n| \neq \infty$, since

$$\lim_{k \rightarrow \infty} |a_{2k-1}| = \lim_{k \rightarrow \infty} 0 = 0$$

- (d) Assume that $0 < R < \infty$ and let $0 < \tilde{R} < \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n (x - 5)^n$.

Then $\tilde{R} = R$.

Solution:

True.

If we substitute $t = x - 5$ we get that

$$\sum_{n=0}^{\infty} a_n (x - 5)^n = \sum_{n=0}^{\infty} a_n t^n$$

We know that the radius of convergence of $\sum_{n=0}^{\infty} a_n x^n$ is R , thus the series converges for every

$$|x - 5| = |t| < R$$

and diverges for every

$$|x - 5| = |t| > R$$

Thus, the radius of convergence of $\sum_{n=0}^{\infty} a_n (x - 5)^n$ is also R and $\tilde{R} = R$.

- (e) It is possible that both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} (-1)^n a_n$ are conditionally convergent.

Solution:

True.

$$\text{We choose } a_0 = 0 \text{ and } a_n = \begin{cases} \frac{(-1)^k}{k} & n = 2k \\ 0 & n = 2k - 1 \end{cases} = \begin{cases} \frac{2(-1)^{\frac{n}{2}}}{n} & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases} \text{ for every } n \in \mathbb{N}.$$

In this case we get that

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{n}$$

We note that for $x = 1$ and $x = -1$ we get that

$$\begin{aligned} \sum_{n=0}^{\infty} a_n &= \sum_{n=1}^{\infty} \frac{(-1)^n 1^{2n}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \\ \sum_{n=0}^{\infty} (-1)^n a_n &= \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^{2n}}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \end{aligned}$$

Thus both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} (-1)^n a_n$ are conditionally convergent.

5. (a) Let $(a_k)_{k=1}^{\infty}$ be a non-negative bounded sequence such that $\sum_{k=1}^{\infty} a_k$ diverges.

Let $(S_n)_{n=1}^{\infty}$ be the sequence of partial sums of $(a_k)_{k=1}^{\infty}$, namely,

$$S_n = \sum_{k=1}^n a_k, \quad \forall n \in \mathbb{N}.$$

Find the domain of convergence of the power series $\sum_{n=1}^{\infty} S_n x^n$.

Solution:

First, as the series $\sum_{k=1}^{\infty} a_k$ is a divergent nonnegative series, then

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \sum_{k=1}^{\infty} a_k = \infty$$

We now show that the power series $\sum_{n=1}^{\infty} S_n x^n$ diverges for $x = \pm 1$:

Note that for such x ,

$$\lim_{n \rightarrow \infty} |S_n \cdot x^n| = \lim_{n \rightarrow \infty} |S_n| = \infty \Rightarrow \lim_{n \rightarrow \infty} [S_n \cdot x^n] \neq 0$$

Thus the power series $\sum_{n=1}^{\infty} S_n x^n$ diverges by the necessary criterion.

It follows that the power series $\sum_{n=1}^{\infty} S_n x^n$ diverges also diverges for every $|x| \geq 1$.

We now prove that the power series converges absolutely for every $|x| < 1$:

Let $|x| < 1$. As $(a_k)_{k=1}^{\infty}$ is bounded and non-negative, there exists an $M \geq 0$ such that

$$0 \leq a_k \leq M, \quad \forall k \in \mathbb{N}$$

It follows that

$$S_n = \sum_{k=1}^n a_k \leq \sum_{k=1}^n M = M \cdot n, \quad \forall n \in \mathbb{N}$$

Consequently,

$$\sum_{n=1}^{\infty} S_n |x|^n \leq \sum_{n=1}^{\infty} M \cdot n \cdot |x|^n = M \sum_{n=1}^{\infty} n \cdot |x|^n$$

Consider the power series $\sum_{n=1}^{\infty} n \cdot x^n$.

Note that its convergence radius is

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{n}} = 1$$

Thus $\sum_{n=1}^{\infty} n \cdot |x|^n < \infty$ and therefore $\sum_{n=1}^{\infty} S_n |x|^n < \infty$.

We conclude that the domain of convergence of the power series $\sum_{n=1}^{\infty} S_n x^n$ is $(-1, 1)$.

(b) Conclude that the domain of convergence of the power series

$$\sum_{n=1}^{\infty} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) x^n \text{ is } (-1, 1).$$

Solution:

Consider the sequence $(a_k)_{k=1}^{\infty}$ defined by $a_k = \frac{1}{k}$ for every $k \in \mathbb{N}$.

Note that for every $k \in \mathbb{N}$,

$$0 \leq a_k = \frac{1}{k} \leq 1$$

Thus $(a_k)_{k=1}^{\infty}$ is non-negative and bounded.

We also have

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \frac{1}{k} \stackrel{\text{Harmonic}}{=} \infty$$

Thus, by item (a), the domain of convergence of

$$\sum_{n=1}^{\infty} S_n x^n = \sum_{n=1}^{\infty} \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) x^n$$

is $(-1, 1)$.

6. Let $x_0 \in \mathbb{R}$ and let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers.

Let $0 < R < \infty$ be the convergence radius of the power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ and let $x_1 \in \mathbb{R}$ such that $|x_1 - x_0| > R$. Prove that $\lim_{n \rightarrow \infty} a_n (x_1 - x_0)^n \neq 0$.

Solution:

Assume towards a contradiction that $\lim_{n \rightarrow \infty} a_n (x_1 - x_0)^n = 0$, then there exists an $N \in \mathbb{N}$ such that for every $n \geq N$ we have

$$|a_n (x_1 - x_0)^n| < \boxed{1} \quad (*)$$

Let $x_2 \in \mathbb{R}$ such that $R < |x_2 - x_0| < |x_1 - x_0|$.

Thus

$$\sum_{n=N}^{\infty} |a_n (x_2 - x_0)^n| = \sum_{n=N}^{\infty} |a_n (x_2 - x_0)^n| \cdot \frac{|x_1 - x_0|^n}{|x_1 - x_0|^n} = \sum_{n=N}^{\infty} |a_n (x_1 - x_0)^n| \cdot \frac{|x_2 - x_0|^n}{|x_1 - x_0|^n} \leq \sum_{n=N}^{\infty} 1 \cdot \left(\frac{|x_2 - x_0|}{|x_1 - x_0|} \right)^n$$

Set $q = \frac{|x_2 - x_0|}{|x_1 - x_0|}$. As $|x_2 - x_0| < |x_1 - x_0|$, then $0 < q < 1$.

Thus $\sum_{n=N}^{\infty} q^n$ is the tail of a convergent geometric series, and therefore

$$\sum_{n=N}^{\infty} |a_n (x_2 - x_0)^n| \leq \sum_{n=N}^{\infty} \left(\frac{|x_2 - x_0|}{|x_1 - x_0|} \right)^n = \sum_{n=N}^{\infty} q^n < \infty$$

It follows that $\sum_{n=0}^{\infty} a_n (x_2 - x_0)^n$ converges absolutely, contradicting the fact that $|x_2 - x_0| > R$.